



HA NOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

Introduction to Cryptography and Security

Perfect Security

Slides are taken from

- <https://cseweb.ucsd.edu/~mihir/cse107/slides.html>

Outline

- 1 Definition
- 2 One-time Pad
- 3 One-Time Pad Security

A measure of security

Let (Enc, Dec) be a symmetric encryption scheme. For any message m and ciphertext c we are interested in

$$\Pr[\text{Enc}(k, m) = c]$$

where the probability is over the random choice $k \leftarrow \mathcal{K}$ and over the coins tossed by Enc if any.

Example

Consider the symmetric encryption scheme as follows.

		messages:			
		00	01	10	11
keys:	00	01	10	11	00
	01	01	11	10	00
	10	00	11	01	11
	11	11	10	01	11

The table entry in row k and column m is $\text{Enc}(k, m)$,

- $\Pr[\text{Enc}(k, 00) = 01] = 2/4 = 1/2$
- $\Pr[\text{Enc}(k, 01) = 01] = 0$
- $\Pr[\text{Enc}(k, 10) = 11] = 1/4$

Perfect Security

Definition

Let $\Pi = (\text{Enc}, \text{Dec})$ be a symmetric encryption scheme. We say that Π is **perfectly secure** if for any two messages m_1, m_2 and any ciphertext c

$$\Pr[\text{Enc}(k, m_1) = c] = \Pr[\text{Enc}(k, m_2) = c].$$

In both cases, the probability is over the random choice $k \leftarrow \mathcal{K}$ and over the coins tossed by **Enc** if any.

Intuitively: Given c , and even knowing the message is either m_1 or m_2 the adversary cannot determine which.

Perfect Security

Definition requires that

For all m_1, m_2, c we have

$$\Pr[\text{Enc}(k, m_1) = c] = \Pr[\text{Enc}(k, m_2) = c].$$

If we want to show the definition is **not** met, we need to show that

There exists m_1, m_2, c such that

$$\Pr[\text{Enc}(k, m_1) = c] \neq \Pr[\text{Enc}(k, m_2) = c].$$

Example

		messages:			
		00	01	10	11
keys:	00	01	10	11	00
	01	01	11	10	00
	10	00	11	01	11
	11	11	10	01	11

The table entry in row k and column m is $\text{Enc}(k, m)$.

- $\Pr[\text{Enc}(k, 00) = 01] = 2/4 = 1/2$
- $\Pr[\text{Enc}(k, 01) = 01] = 0$

Question: Is this encryption scheme perfectly secure? **No**, because for $m_1 = 00, m_2 = 01$ and $c = 01$ we have

$$\Pr[\text{Enc}(k, m_1) = c] \neq \Pr[\text{Enc}(k, m_2) = c].$$

Perfect security of substitution ciphers

Claim

*A substitution cipher is **NOT** perfectly secure.*

Example

$A \rightarrow k$

$B \rightarrow d$

$C \rightarrow w$

...

Perfect security of substitution ciphers

Claim

Let $\Pi = (\text{Enc}, \text{Dec})$ be a substitution cipher over the alphabet Σ consisting of the 26 English letters. Assume that k picks a random permutation over Σ as the key. That is, its code is

$$k \leftarrow \text{PERM}(\Sigma); \quad \text{return } k.$$

Let *Plaintexts* be the set of all three letter English words. Then Π is not perfectly secure.

Proof of claim

To show: there exist m_1, m_2, c such that

$$\Pr[\text{Enc}(k, m_1) = c] \neq \Pr[\text{Enc}(k, m_2) = c].$$

Let

- $c = \text{xyy}$
- $m_1 = \text{FEE}$
- $m_2 = \text{FAR}$

Then

$$\begin{aligned}\Pr[\text{Enc}(k, m_2) = c] &= \Pr[\text{Enc}(k, \text{FAR}) = \text{xyy}] \\ &= 0\end{aligned}$$

Why?

Proof of claim

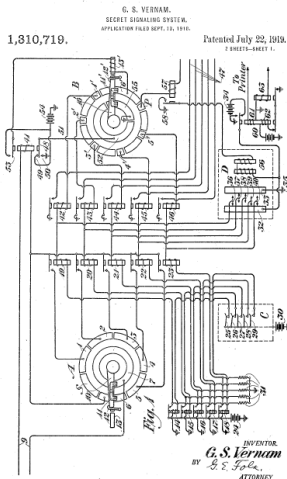
$$\begin{aligned}\Pr[\text{Enc}(k, m_1) = c] &= \Pr[\text{Enc}(k, \text{FEE}) = \text{xyy}] \\ &= \frac{|\{k \in \text{PERM}(\Sigma) : k(\text{F})k(\text{E})k(\text{E}) = \text{xyy}\}|}{|\text{PERM}(\Sigma)|} \\ &= \frac{24!}{26!} \\ &= \frac{1}{650}.\end{aligned}$$

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One-Time Pad or OTP

- Vernam 1917. Paper-tape based (patent)
- Message, key, ciphertext have the same length (λ bit).
- Key k also called pad; it is random and known only by Alice & Bob.



XOR Operation

XOR of two strings in $\{0, 1\}^n$ is their bit-wise addition mod 2.

x	y	$\oplus$
0	0	0
0	1	1
1	0	1
1	1	0

$$\begin{array}{r} 1\ 0\ 1\ 1\ 0\ 0 \\ \oplus\ 0\ 1\ 1\ 0\ 1\ 0 \\ \hline 1\ 1\ 0\ 1\ 1\ 0 \end{array}$$

An important property of XOR

Theorem

Let x a random variable $\{0, 1\}^n$, and k an independent uniform variable on $\{0, 1\}^n$.

Then $y = x \oplus k$ is uniform variable on $\{0, 1\}^n$.

Proof.

For $n = 1$, we have:

x	Pr
0	p_0
1	p_1

k	Pr
0	$1/2$
1	$1/2$

x	k	Pr
0	0	$p_0/2$
0	1	$p_0/2$
1	0	$p_1/2$
1	1	$p_1/2$



One Time Pad

- **Gen:** Generates a random bit sequence of length λ .
- **Enc:** Represent the message as a binary string and XOR with the key.

$$\begin{array}{rcl} x & = & 101100.. \\ \oplus \quad k & = & 011010.. \\ \hline y & = & 110110.. \end{array}$$

- **Dec:** Same as encryption, just XOR with k .

$$\begin{aligned} (x_i \oplus k_i) \oplus k_i &= x_i \oplus (k_i \oplus k_i) \\ &= x_i \oplus 0 = x_i \end{aligned}$$

Exercise

List the advantages and disadvantages of OTP.

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One Time Pad

- **Gen:** Generates a random bit sequence of length λ .
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Intuition for OTP security

Suppose adversary gets ciphertext $c = 101$ and knows the plaintext m is either $m_1 = 010$ or $m_2 = 001$. Can it tell which?

No, because $c = k \oplus m$ so

- $m = 010$ iff $k = 111$
- $m = 001$ iff $k = 100$

but k is equally likely to be 111 or 100 and adversary does not know k .

Perfect security of OTP

Claim

Let $\Pi = (\text{Enc}, \text{Dec})$ be the OTP scheme with key-length $\lambda \geq 1$. Then Π is perfectly secure.

Proof Idea.

Want to show that for any m_1, m_2, c

$$\Pr[\text{Enc}(k, m_1) = c] = \Pr[\text{Enc}(k, m_2) = c].$$

That is

$$\Pr[k \oplus m_1 = c] = \Pr[k \oplus m_2 = c]$$

when $k \leftarrow \{0, 1\}^\lambda$.



Example: $\lambda = 2$

		messages:			
keys:		00	01	10	11
	00	00	01	10	11
	01	01	00	11	10
	10	10	11	00	01
	11	11	10	01	00

The table entry in row k and column m is $\text{Enc}(k, m) = k \oplus m$.

- $\Pr[\text{Enc}(k, 00) = 01] = 1/4$
- $\Pr[\text{Enc}(k, 10) = 01] = 1/4$

Proof of Claim

$$\begin{aligned}\Pr[\text{Enc}(k, m) = c] &= \Pr[k \oplus m = c] \\ &= \frac{|\{k \in \{0, 1\}^\lambda : k \oplus m = c\}|}{|\{0, 1\}^\lambda|} \\ &= 1/2^\lambda.\end{aligned}$$

Perfect security: Plusses and Minuses

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- Very good privacy

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- Key needs to be as long as message

Project 1: Many-time pad attack

<https://www.coursera.org/learn/crypto/>

- Let us see what goes wrong when an OTP key is used more than once.
- Given eleven hex-encoded ciphertexts that are the result of encrypting eleven plaintexts with an OTP scheme, all with the same OTP key.
- Your goal is to decrypt the last ciphertext, and submit the secret message within it as solution.



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Thank you!



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